CS4061D Topics in Data Analytics

Fine Tuning Gemma for Text-to-SQL conversion

Abstract:

Natural language interfaces to databases remain a central challenge in human–computer interaction, as users often lack expertise in structured query languages. Tasks such as **Knowledge Graph Completion (KGC**) highlight the need to translate natural language into SQL queries effectively. This research investigates the fine-tuning of Google's **Gemma** language model for **text-to-SQL** conversion using parameter-efficient methods. We propose a multi-stage pipeline that leverages **Quantized Low-Rank Adaptation (QLoRA)** within the **Hugging Face** ecosystem to enable efficient training on resource-constrained hardware. By adapting Gemma through lightweight tuning, the approach ensures scalability and accessibility while preserving general model capabilities. The resulting workflow aims to streamline database query generation, reduce human intervention, and enhance usability in real-world data management systems.

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